

Stabilization of a class of nonlinear systems by a smooth feedback control

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We consider the stabilizability problem for systems of the form $\dot{x} = f(x) + ub$ in the neighborhood of an equilibrium point of $f(x)$. First, by means of center manifold theory, a lower order system is introduced. If this system is stabilizable, then so is the original system. Results on the stabilization of the low order system are presented. No resort is taken to Liapunov theory. The relation between stabilizability and controllability is discussed.

Keywords: Nonlinear system, Stabilizability, Nonlinear feedback, Center manifold, Hopf bifurcation.

1. Introduction

We consider the local behavior of control systems described by

$$\dot{x} = f(x) + ub$$

in the neighborhood of an equilibrium point x_e of $f(x)$. Without loss of generality x_e is assumed to be the origin. In a recent paper [1] we have established sufficient conditions for local and global controllability of such systems. Our main goal here is to construct – if possible – a smooth function $u(x)$ such that the origin is an equilibrium point which is asymptotically stable. Sections 2 and 3 are devoted to this problem. In Section 4 we briefly discuss the relation between controllability and stabilizability by smooth feedback.

Stabilization by smooth feedback has been investigated by Brockett [2]. His theory rests on Liapunov functions, in order to check what he calls the ‘finite gain property’ of a derived system (for more details, see [2]). We are taking a different approach. Our results are based on center manifold theory [3], which enables us to reduce the problem to one of lower dimension. The stability

properties of this derived system are then investigated. We will give – as a matter of illustration – a class of systems for which this program can be carried out successfully.

Probably the main advantage of this approach is that it is entirely independent of one’s ability to construct an appropriate Liapunov function. In addition the reduction procedure is independent of the order of the original system.

2. Reduction to a system of lower order

We will consider systems of the form

$$\dot{x} = f(x) + ub \quad (2.1)$$

with scalar input u , $f(0) = 0$ and with constant control vector b . Generalizations of the approach to more general systems $\dot{x} = f(x, u)$ are straightforward.

First, linear system theory is invoked to a point where nonlinear arguments are needed in order to obtain additional results. Consider

$$\dot{x} = Ax + bu + O(2)$$

where $A = (\partial f / \partial x)(0)$ and where $O(2)$ are the higher order terms of $f(x)$. By a change of basis (2.1) is transformed into

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} b_1 \\ 0 \end{pmatrix} u + O(2)$$

with (A_{11}, b_1) a controllable pair. It follows from linear system theory that there exists a linear feedback control $u = k_1 x_1$ such that $A_{11} + b_1 k_1$ is Hurwitz. Applying $u = k_1 x_1 + v$ we obtain

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} A_{11} + b_1 k_1 & A_{12} \\ 0 & A_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} b_1 \\ 0 \end{pmatrix} v + O(2). \quad (2.2)$$

We conclude that (2.1) is stabilizable by linear feedback if the *uncontrollable modes* (correspond-

ing to the eigenvalues of A_{22}) are asymptotically stable. On the other hand, if A_{22} has eigenvalues with positive real part, it is well known that the system $\dot{x} = f(x) + ub$ cannot be stabilized at all.

From these remarks we see that in order to have a solvable and nontrivial problem we must assume that A_{22} has eigenvalues with real part ≤ 0 ; those eigenvalues with real part equal to zero are called critical.

By an appropriate choice of k_1 we can assume that the eigenvalues of $A_{11} + b_1 k_1$ are different from those of A_{22} . Then, after performing another change of basis so as to block-diagonalize the linear part of (2.2) we arrive at (after a change of notation)

$$\begin{aligned}\dot{x}_1 &= A_{11}x_1 + f_1(x_1, x_2, x_3), \\ \dot{x}_2 &= A_{22}x_2 + f_2(x_1, x_2, x_3), \\ \dot{x}_3 &= A_{33}x_3 + f_3(x_1, x_2, x_3) + bu,\end{aligned}\quad (2.3)$$

with A_{11} critical, A_{22} and A_{33} having eigenvalues with negative real part and A_{33} such that (A_{33}, b) is a controllable pair. The functions f_1 , f_2 and f_3 are the higher order terms; they vanish at the origin.

The problem has been reduced to the construction of a feedback control law $u = F(x_1, x_2, x_3)$ such that the origin of (2.3) is asymptotically stable.

Consider the (generalized) eigenspace E^c corresponding to the eigenvalues of A_{11} . Then, by the center manifold theorem [3,p.4] there exists a 'center manifold' W^c tangent to E^c at the origin. This center manifold has sufficiently high degree of smoothness and is invariant with respect to the flow.

Since W^c is tangent to E^c (the space $x_2 = x_3 = 0$) it can be represented as a (local) graph in the (x_1, x_2, x_3) -space by

$$W^c = \{(x_1, h(x_1)); h(0) = Dh(0) = 0\}$$

with $h = (h_1, h_2)$ defined in some neighborhood of the origin. Consider the projection of the vector field on W^c onto E^c ,

$$\dot{x}_1 = A_{11}x_1 + f_1(x_1, h_1(x_1), h_2(x_1)). \quad (2.4)$$

Since $h(x_1)$ is tangent to $x_2 = x_3 = 0$ and since the $\dot{x}_2$ and $\dot{x}_3$ equations are asymptotically stable up to first order, it is reasonable to expect that the stability behavior of (2.3) is determined by that of

(2.4). In fact this has been shown to be true. We state this as a theorem; for a proof we refer to the literature.

Theorem 1 [3, p. 4]. *If the origin of (2.4) is locally asymptotically stable, then the origin of (2.3) is also locally asymptotically stable.*

We now show how $h(x_1)$ can be calculated, or at least approximated. Substituting x_2 by $h_1(x_1)$ and x_3 by $h_2(x_1)$, by the invariance of the flow on $h(x_1)$ we require

$$\begin{pmatrix} \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = Dh(x_1) \cdot \dot{x}_1.$$

Then h is the solution of the partial differential equation

$$\begin{aligned}Dh(x_1)(A_{11}x_1 + f_1(x_1, h_1(x_1), h_2(x_1))) \\ = \begin{pmatrix} A_{22}h_1(x_1) + f_2(x_1, h_1(x_1), h_2(x_1)) \\ A_{33}h_2(x_1) + f_3(x_1, h_1(x_1), h_2(x_1)) \\ + bF(x_1, h_1(x_1), h_2(x_1)) \end{pmatrix} \end{aligned} \quad (\text{PDE})$$

with boundary conditions

$$h(0) = Dh(0) = 0.$$

By the following theorem the solution $h(x_1)$ can be approximated arbitrarily closely as a Taylor series at $x_1 = 0$.

Theorem 2 [3, p. 5]. *If a function $\phi(x_1)$ ($\phi = (\phi_1, \phi_2)$) exists with $\phi(0) = D\phi(0) = 0$, which satisfies (PDE) modulo terms of the p -th degree and higher ($p > 1$), then*

$$h_1(x_1) = \phi_1(x_1) + O(|x_1|^p),$$

$$h_2(x_1) = \phi_2(x_1) + O(|x_1|^p).$$

By Theorem 1, for (2.3) to be asymptotically stable we need to establish asymptotic stability of the reduced system

$$\begin{aligned}\dot{x}_1 &= A_{11}x_1 \\ &\quad + f_1(x_1, \phi_1(x_1) + O(|x_1|^p), \phi_2(x_1) + O(|x_1|^p))\end{aligned}$$

From now on, assume that the reduced system is asymptotically stable if and only if

$$\dot{x}_1 = A_{11}x_1 + f_1(x_1, \phi_1(x_1), \phi_2(x_1)) \quad (2.5)$$

is asymptotically stable. This needs justification in general.

Recalling that ϕ_1 and ϕ_2 depend on $u = F(x_1, x_2, x_3)$ through the partial differential equation which determines the center manifold, our task is to construct $F(x_1, x_2, x_3)$ such that (2.5) is asymptotically stable. Our ability to do so will of course depend on the structure of f_1 , f_2 and f_3 .

We will not discuss the *scalar* x_1 case (i.e. $A_{11} = 0$) in general, since it is rather trivial. Simple examples are given in the remarks at the end of Section 3 which are meant, among other things, to illustrate this case. We are mainly interested in the case where x_1 is two-dimensional and especially when A_{11} has two purely imaginary eigenvalues.

3. Construction of a stabilizing control law

In the previous section we have been mainly concerned with the reduction of the original stabilization problem to a problem of lower dimension (equal to the dimension of x_1). It is needless to say that if $\dim x_1 > 1$, the stability investigation of the reduced equation can be quite involved. In this section we will show that if $\dim x_1 = 2$ and A_{11} has two purely imaginary eigenvalues some general statements can be made on the stabilizability of such systems. Other cases (A_{11} having eigenvalues equal to zero and diagonalizable or not) need further investigation and should probably be studied within the context of singularity theory [4].

Consider the second order system

$$\ddot{x} + x - g(x, \dot{x}, y) = 0$$

with $x, \dot{x}, y \in \mathbb{R}$. The function $g(x, \dot{x}, y)$ contains only terms of order ≥ 2 (in all variables). The parameter y will be considered as the output of a controllable linear system which we assume scalar for explanatory convenience. More generally, y could be the (nonlinear) output $v(z)$ of a nonlinear control system $\dot{z} = w(z, u)$ of arbitrary dimension such that its linearization at $(z, u) = (0, 0)$ is a controllable system.

In (standard) state space coordinates the control system is written as

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ g(x_1, x_2, y) \end{pmatrix}, \quad (3.1)$$

$$\dot{y} = ly + u,$$

which control u . The problem is to construct a feedback $u = F(x_1, x_2, y)$ such that the overall system is locally asymptotically stable. Our ability to do so will depend on the function $g(x_1, x_2, y)$. Notice that the $\dot{x}_1$ -equation contains no higher order terms. It will be clear from what follows that our results can be adapted to the more general case, where higher order terms appear in the $\dot{x}_1$ -equation.

Setting $u = -ly - ky + w$, $k > 0$, yields

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ g(x_1, x_2, y) \end{pmatrix}, \quad (3.2)$$

$$\dot{y} = -ky + w.$$

This system is of the form described in Section 2: it has block-diagonal form with the critical modes noncontrollable. Let w be in feedback form $w(x_1, x_2, y)$, only containing terms of order ≥ 2 (in all variables). This is a restriction to which we will return in the remarks.

Let

$$\begin{aligned} y &= h(x_1, x_2) \\ &= ax_1^2 + bx_1x_2 + cx_2^2 + O(3) \end{aligned}$$

be the center manifold of the system under study, i.e.

$$\begin{aligned} Dh(x) \begin{pmatrix} x_2 \\ -x_1 + g(x_1, x_2, h(x_1, x_2)) \end{pmatrix} \\ + kh(x_1, x_2) - w(x_1, x_2, h(x_1, x_2)) = 0. \end{aligned}$$

Taking

$$w(x_1, x_2, y) = \alpha x_1^2 + \beta x_1x_2 + \gamma x_2^2,$$

then after substitution of the expressions for w and h in the equation above and setting the lowest order terms (i.e. second order) equal to zero, it follows that $\alpha x_1^2 + \beta x_1x_2 + \gamma x_2^2$ is a second order approximation to the center manifold if

$$2a - 2c + kb - \beta = 0,$$

$$-b + ka - \alpha = 0,$$

$$b + kc - \gamma = 0.$$

Since $k \neq 0$, a , b and c can be assigned arbitrary values by adjusting the (feedback) parameters α , β , γ . In other words, appropriate nonlinear feedback determines the lowest order terms of the center manifold.

By Theorem 1, system (3.2) is locally asymptotically stable if

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ g(x_1, x_2, ax_1^2 + bx_1x_2 + cx_2^2 + O(3)) \end{pmatrix} \quad (3.3)$$

is locally asymptotically stable.

Theorem 3. *If g contains terms of the form x_1y and/or x_2y , then the system (3.1) is stabilizable by feedback (i.e. there exists a smooth feedback control $u(x_1, x_2, y)$ such that the overall system is asymptotically stable at the origin).*

Proof. As explained earlier, system (3.1) is stabilizable, if (3.3) is asymptotically stable. By the normal form theorem [5] there exist nonlinear transformations which bring the 3-jet of (3.3) (i.e. the Taylor series expansion up to degree 3) to the following form (in polar coordinates):

$$\begin{aligned} \dot{r} &= dr^3, \\ \dot{\theta} &= f(r). \end{aligned} \quad (3.4)$$

By the Hopf bifurcation theorem [5], the qualitative properties of (3.4) remain unchanged if higher order terms are added to (3.4) – assuming that $d \neq 0$. In particular, approximating the center manifold to third or higher degree is not necessary, when one investigates the stability behavior of (3.1). In other words d is determined by the second and third order terms of (3.3). If $d < 0$ the system (3.4) is asymptotically stable and stabilizability of (3.1) follows.

All that remains is to derive an expression for d as a function of g . We again refer to [5] where an expression for d has been obtained. For the problem at hand

$$d = \frac{1}{16} [g_{x_1x_1x_2} + g_{x_2x_2x_2}] + \frac{1}{16} [g_{x_1x_2}(g_{x_1x_1} + g_{x_2x_2})]$$

where indices x_i mean partial derivatives with respect to x_i (only the second and third order terms of g are taken into account).

Assume that g contains the term x_1y ; then after substitution of $y = h(x_1, x_2)$ the expression for d carries terms of the form

$$ax_1^3 + bx_1^2x_2 + cx_1x_2^2.$$

Therefore by picking b in an appropriate manner d can be assigned negative values. A similar argument applies when the term x_2y is present in $g(x_1, x_2, y)$. Then a and (or) c can be chosen such that d is negative.

Remarks and more examples. (1) For the class of systems discussed above it can be shown along the same lines, that the system can be stabilized by linear feedback. The calculations however turn out to be more involved since a block-diagonalizing transformation is needed first. As a consequence, the $\dot{x}_1$ -equation contains higher order terms which from a computational viewpoint lengthens the analysis.

(2) As a matter of completeness we give an example of a system which *cannot be stabilized by linear feedback but is stabilizable by nonlinear feedback*. At the same time, this example serves as an illustration for the case where the reduced equation is *scalar*. Consider

$$\dot{x} = xy,$$

$$\dot{y} = u.$$

The feedback $u = -y + \alpha x^2$, $\alpha < 0$, stabilizes the system. The center manifold is given by $y = h(x)$ with $h(x) = \alpha x^2 + O(x^4)$. Indeed, with the proposed feedback the differential equation determining h is

$$h'(x)[xh(x)] + h(x) - \alpha x^2 = 0.$$

Letting $h = \alpha x^2 + bx^3 + \dots$, one obtains

$$h = \alpha x^2 + O(x^4).$$

The reduced system is

$$\dot{x} = \alpha x^3 + O(x^5)$$

which is asymptotically stable. This implies asymptotic stability of the original system.

The system above is *not* stabilizable by linear feedback. Indeed, without loss of generality we can assume $u = -y + \beta x$. Then

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ \beta & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} xy \\ 0 \end{pmatrix}.$$

After a diagonalizing transformation

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \beta & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

one obtains

$$\begin{pmatrix} \dot{x}_1 \\ \dot{y}_1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_1(\beta x_1 + y_1) \\ -\beta x_1(\beta x_1 + y_1) \end{pmatrix}.$$

This system falls within the framework set up in Section 2 (the linear part is in diagonal form and has an eigenvalue equal to zero and a stable eigenvalue). Therefore the stability behavior of this system is given by the stability behavior of the reduced system

$$\dot{x}_1 = x_1(\beta x_1 + y_1)$$

with $y_1 = h(x_1)$ describing the center manifold. But $h(x_1)$ is of degree at least 2, thus the reduced system is of the form

$$\dot{x}_1 = \beta x_1^2 + O(x_1^3)$$

which is *unstable* independent of β . This proves the assertion.

(3) As another example of a system with a *single* critical uncontrollable mode, consider the system studied by Brockett [2]

$$\dot{z} = xy,$$

$$\dot{x} = u,$$

$$\dot{y} = v,$$

which describes the angular velocity equations of a rigid body controlled by *two* gas jets (modeled by the inputs u and v) along the principal axes.

Consider the feedback law

$$u = -x + \alpha z^2,$$

$$v = -y + \beta z^3.$$

Then

$$\begin{pmatrix} \dot{z} \\ \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} z \\ x \\ y \end{pmatrix} + \begin{pmatrix} xy \\ \alpha z^2 \\ \beta z^3 \end{pmatrix}$$

which has a linear part in diagonal form with a zero eigenvalue and two eigenvalues with negative real part. Therefore there exists a center manifold described by

$$x = h_1(z),$$

$$y = h_2(z),$$

which has the property that the stability behavior of the system above is given by the stability behavior of the reduced system

$$\dot{z} = h_1(z)h_2(z).$$

Let

$$h_1(z) = h_{11}z^2 + h_{12}z^3 + \dots,$$

$$h_2(z) = h_{21}z^2 + h_{22}z^3 + \dots.$$

From the differential equation which determines (h_1, h_2) and from Theorem 2 (see Section 2) it follows that

$$h_{11} = \alpha, \quad h_{21} = 0, \quad h_{22} = \beta,$$

which gives for the reduced equation

$$\dot{z} = \alpha\beta z^5 + \text{higher order terms.}$$

This system is locally asymptotically stable if $\alpha\beta < 0$. Therefore

$$u = -x + \alpha z^2,$$

$$v = -y + \beta z^3,$$

with $\alpha\beta < 0$ is a stabilizing feedback.

It is also easy to recover the stabilizing control

$$u = -x + z,$$

$$v = -y - z^2$$

proposed by Brockett [2]. Indeed, with this control the system becomes

$$\begin{pmatrix} \dot{z} \\ \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} z \\ x \\ y \end{pmatrix} + \begin{pmatrix} xy \\ 0 \\ -z^2 \end{pmatrix}.$$

Performing a diagonalizing linear transformation

$$\begin{pmatrix} z \\ x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

it follows that

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} (x_1 + x_2)x_3 \\ -(x_1 + x_2)x_3 \\ -x_1^2 \end{pmatrix}$$

which is a system having the properties required by Section 2. Therefore there exists a center manifold described by

$$x_2 = h_1(x_1) = h_{11}x_1^2 + h_{12}x_1^3 + \dots,$$

$$x_3 = h_2(x_1) = h_{21}x_1^2 + h_{22}x_1^3 + \dots.$$

From the theory of Section 2 it follows that

$$h_{11} = 0, \quad h_{21} = -1.$$

The reduced equation is

$$\dot{x}_1 = (x_1 + 0 \cdot x_1^2 + \dots)(-x_1^2 + \dots)$$

or

$$\dot{x}_1 = -x_1^3 + \text{higher order terms,}$$

which is locally asymptotically stable at the origin.

(4) As explained in Section 2, we have assumed that the linearized system involves critical uncontrollable modes in order to have a nontrivial stabilization problem. Consider the case where there are no critical uncontrollable modes, but instead modes with a small negative real part besides possibly uncontrollable modes with large negative real part. This phenomenon could be brought about by introducing a small linear damping term in the second order equation of Section 3. The system with no control is then asymptotically stable, due to this *weak damping effect* and a stabilizing feedback is redundant as far as asymptotic stability is concerned. However, consider the system with weak damping and apply the theory above to construct a stabilizing feedback law for the approximate system obtained after neglecting the weak damping terms. It seems plausible to expect that this stabilizing feedback applied to the original system (with weak damping) accelerates the convergence to the origin if compared to the original system with no control. Further investigation in this direction is necessary.

4. Controllability and stabilizability

In this final section we present a *negative result*. We briefly discuss an example for which controllability does *not* imply stabilization by *smooth* feedback, as it does for linear systems. This shows that the concept of controllability in a nonlinear context is less relevant in the feedback stabilization problem (unless one allows discontinuous feedback and 'realizes' the open loop control as a feedback control [6]). This seems to suggest that one must tackle the feedback stabilization problem

directly (i.e. more or less independent of the controllability properties). The techniques discussed above and in [2] serve as possible approaches.

Consider the system

$$\dot{x} = x + y^3,$$

$$\dot{y} = f(x, y) + u,$$

with $f(0, 0) = 0$ and u a scalar control. This system is locally controllable at the origin [1] (if there are no bounds on u , it is actually globally controllable [1]). Notice that the linearization at the origin has a positive eigenvalue ($= 1$), independent of the feedback $u(x, y)$. This means that the system cannot be stabilized by a *smooth feedback*.

This is not the first example showing that controllability does not necessarily imply the existence of a smooth stabilizing feedback control. In fact, Sussmann [6] and Brockett [2] have given examples. Brockett indicates that his example of a system in $\mathbb{R}^3$ does not involve bounds on the controls and nonlocal considerations as Sussmann's example does. Our example has the same characteristics with respect to Sussmann's example. It is in our opinion conceptually simpler than Brockett's example since – besides being a system in the plane – the obstruction to stabilization follows from the presence of an *uncontrollable unstable* mode.

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